DEEP LEARNING CIE SUBMISSION

PBL TOPIC: ATTENTION-BASED TEXT CLASSIFICATION

Student Name:	Abhijit Kamasane
Roll No:	34131
Project Topic:	Attention-Based Text Classification
Paper Reviewed:	*Neural Machine Translation by Jointly Learning to Align and Translate*
Authors & Year:	Dzmitry Bahdanau, Kyunghyun Cho, Yoshua Bengio (2014)
Publication Venue:	arXiv preprint | arXiv:1409.0473

1. Problem Addressed

Addresses the severe information bottleneck of conventional sequence-to-sequence architectures, where a long input sentence is forced into a fixed-length vector.

2. Motivation & Need

Standard encoder-decoder models struggle to translate long sentences because early tokens fade from memory during encoding.

3. Methodology & Model Architecture

Introduced additive attention, allowing the decoder to search over all encoder hidden states h_i to form a dynamic context vector c_i at each decoding step.

4. Dataset & Experimental Setup

WMT 2014 English-to-French translation task comprising millions of sentence pairs.

5. Attention Mechanism Formulation

Score equation: score_i = tanh(W_a * s_{t-1} + U_a * h_i). Softmax weights α_i determine the relative alignment between decoder state s_{t-1} and encoder hidden state h_i.

6. Key Results Reported in Paper

Achieved superior BLEU scores on long sentences (30+ words) compared to standard fixed-vector seq2seq models.

7. System Advantages & Strengths

Eliminates single-vector sequence bottlenecks and allows dynamic token alignment across arbitrary sequence lengths.

8. Limitations & Constraints

Increases quadratic computational complexity due to calculating attention scores across all encoder hidden states at every step.

9. Identified Research Gap

Traditional models lacked dynamic alignment mechanisms for sequence representations.

10. Relevance to Our PBL Project

Provides the foundational additive attention formulation (score = tanh(W_h * h_i + b)) used in our Attention-BiLSTM model.

11. Student's Critical Observations

Bahdanau attention proved that soft-alignment is superior to hard-alignment, laying the groundwork for modern NLP attention networks.

12. References

[1] D. Bahdanau et al., 'Neural machine translation by jointly learning to align and translate,' arXiv:1409.0473, 2014.